Project summary: robustness of LLM-screened priority signals in AMR research

Hyam Ali · September 2026

Over the summer we built a resumable pipeline that pulls the PubMed literature for the 89 WHO GLASS pathogen–antibiotic pairs (11 pathogens and 30 antibiotics, 2000–2026) and asks a language model whether each abstract is about that pair's resistance. The corpus contains 38,987 unique abstracts, which make up 73,857 paper–pair screens. We used Claude Sonnet 5 as the reference and tried 11 other models in 17 model–temperature configurations. Five of these candidates (Gemini 3.5 Flash-Lite, GPT-4.1-nano, GPT-3.5-turbo, Gemma 3 4B and Gemma 3 270M) screened the whole corpus, and six small open-weight configurations formed a size ladder on three representative pairs, giving 711,925 screening decisions in total. For every model, we turned the verdicts into a Scientific Interest signal and a five-year priority forecast under two acceptance rules (strict-only and strict-or-yes). We then compared the results with the reference both per paper (Cohen's κ and macro-F1) and per pair (Spearman ρ, quadrant changes, the ± 1 SD band and overlap in emerging pairs), checked them against a no-LLM baseline that accepts every abstract, and worked out the cost per paper at list prices.

The main conclusion is that whether a cheap model can replace the reference depends more on the acceptance rule than on the model itself. Under strict-or-yes, the candidates disagree with each other on only 11–16% of pair quadrants and rank pairs at $\rho = 0.83$–0.97 against the reference, but the no-LLM baseline already reaches $\rho = 0.95$, so most of that agreement reflects publication volume rather than shared judgement. Under strict-only, where the reference accepts just 23% of abstracts, the candidates disagree on 22–52% of quadrants and the choice of model becomes decisive. Gemini 3.5 Flash-Lite, the only model with moderate per-paper agreement ($\kappa = 0.58$), clearly beats the baseline ($\rho = 0.87$ against 0.67), yet it still moves 38% of pairs into a different quadrant and recovers only 2 of the 13 emerging pairs. We also found that agreement on individual papers and agreement on the overall ranking are separate properties, since Gemma 3 270M agrees at chance per paper but ranks pairs at $\rho = 0.83$, and that small models break down in ways specific to their family. Gemini 3.5 Flash-Lite offers the best trade-off, at \$0.28 per 1,000 papers against \$1.72 for the reference (16% of the cost). The main gaps before submission are a human-labelled validation sample and a re-run of the 6.6% of reference screens that came back empty.